

SVA implication timeline

You have built directed testbenches, classes, constraints, and coverpoints

Overlap, next cycle, and vacuous

- Here is the starter picture
- Antecedent a and consequent b both high at cycle two under overlapping implication
- Shift b one cycle later and overlapping fails while non-overlapping can pass
- If a never goes high
- The lab presets walk overlap pass, overlap fail, next-cycle pass, next-cycle fail
- Rebuild after you change operator or waves so the verdict panel updates before you trust

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / SVA implication timeline

INTERACTIVE TOOL

SVA implication timeline

Sketch a $| \rightarrow$ b (overlap) and a $| \Rightarrow$ b (next-cycle) on a short wave — pass, fail, or **vacuous** success. Not a full SVA engine. Starter: overlapping pass at cycle 2.

Starter example: a $| \rightarrow$ b overlapping pass — at cycle 2 both a and b are high.

Load starter example

Challenges 0 / 22 cleared

Quiz: $| \Rightarrow$: a $| \rightarrow$ b means...

☐ when a is true, b must be true in the same cycle

☐ b is checked two cycles later always

☐ a and b are wires in the DUT netlist only

☐ the assertion is disabled

Show hint

Check

Next

Idle

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

Core ideas

$| \rightarrow$

Overlapping: check b in the same cycle as a.

$| \Rightarrow$

Non-overlap: check b in the next cycle.

Attempt

Each cycle where a is true starts a check.

Vacuous

If a never rises, success is vacuous.

Browser lab starter

Real shell practice

- In the real shell track, open this module's examples prompts
- Restate the timeline idea in one sentence
- Sketch one worked example on paper
- Optional stretch
- No full simulator run required yet, the goal is to read SVA timing in your head

Pitfalls to watch

- Do not treat overlapping and next-cycle operators as interchangeable
- Do not celebrate vacuous green as a design win
- Do not confuse this sketch with full SVA sequences and repetitions
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load starter, run the overlap pass and vacuous presets
- On paper, draw one eight-cycle wave and mark where each operator would pass or fail
- When you are ready, take the short quiz, then continue to the testbench versus UVM map